Plant Growth and Development

NEET-UG Botany | Class 11, Chapter 13

An NCERT-Centric Module for Academic Excellence

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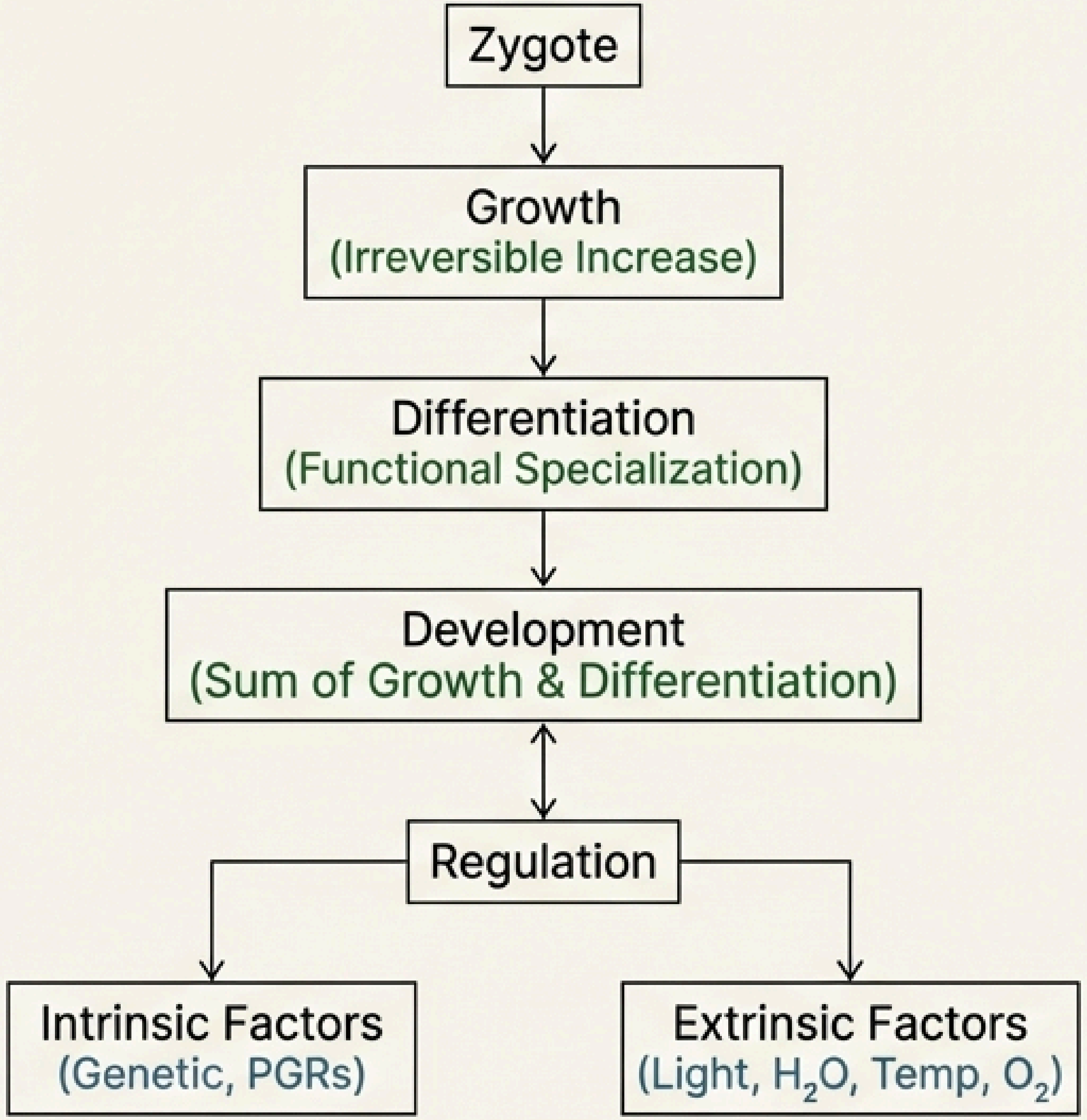
NCERT Syllabus Mapping

- 13.1 Growth
- 13.2 Differentiation, Dedifferentiation and Redifferentiation
- 13.3 Development
- 13.4 Plant Growth Regulators

High-Level Flow of Concepts



Weightage Trend in NEET

High

Analysis: Consistently, **3-4 questions** appear from this chapter each year. The majority of questions are focused on the physiological effects and applications of **Plant Growth Regulators (PGRs)**.

Concept: Growth - Fundamentals (13.1, 13.1.1, 13.1.2)

Key Definitions

Growth: An irreversible permanent increase in size of an organ or its parts or even of an individual cell. It is accompanied by metabolic processes and requires energy.

Indeterminate Growth: Plants retain the capacity for unlimited growth throughout their life due to the presence of **meristems** (e.g., root apical, shoot apical).

Open Form of Growth: New cells are always being added to the plant body by the activity of the meristem.

Growth is Measurable

Growth is measured by parameters proportional to the increase in protoplasm:

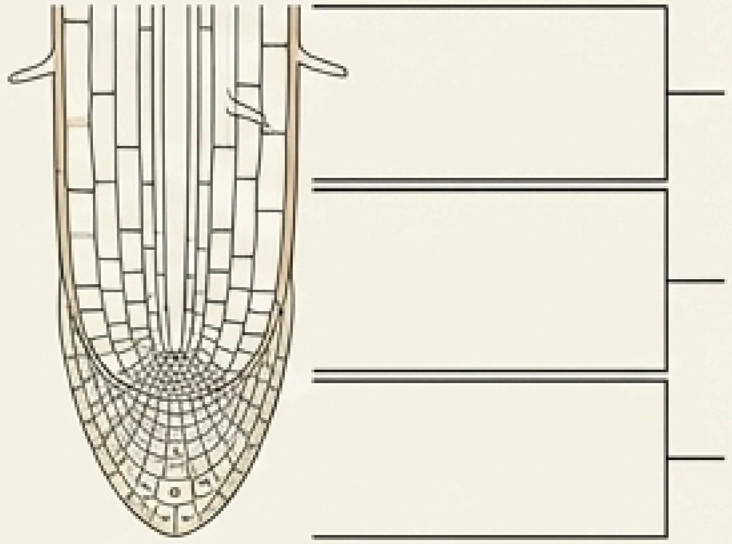
- Increase in fresh weight, dry weight, length, area, volume, and cell number.

Important NCERT Examples

- **Cell Number:** A single maize root apical meristem can produce more than 17,500 new cells per hour.
- **Cell Size:** Cells in a watermelon may increase in size by up to 3,50,000 times.
- **Length:** Growth of a pollen tube.
- **Surface Area:** Growth in a dorsiventral leaf.

Concept: Phases & Rates of Growth (13.1.3, 13.1.4)

Phases of Growth (as seen in root tips):



Maturation Phase
Cells attain maximal size with wall thickening and protoplasmic modifications to perform specific functions.

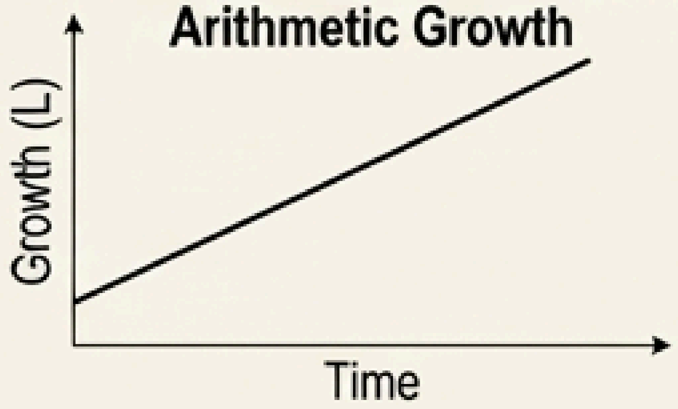
Elongation Phase
Proximal to the meristematic zone, characterized by increased vacuolation, cell enlargement, and new cell wall deposition.

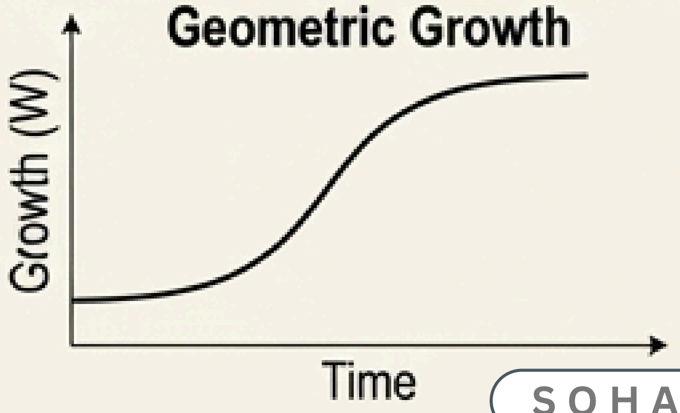
Meristematic Phase
Constantly dividing cells at the apex, rich in protoplasm, with large nuclei and primary cell walls.

The increased growth per unit time.

Types of Growth Rate

Parameter	Arithmetic Growth	Geometric Growth
Cell Division	Only one daughter cell continues to divide; the other differentiates.	Both progeny cells retain the ability to divide.
Curve Shape	Linear Curve	Sigmoid (S-Curve)
Phases	N/A	Lag Phase → Exponential (Log) Phase → Stationary Phase
Formula	$L_T = L_0 + rt$ $W_1 = W_0e^{rt}$ (r is the relative growth rate or efficiency index)	$W_1 = W_0e^{rt}$ (r is the relative growth rate or efficiency index) A living organism growing in a natural environment with limited resources.
Example	A root elongating at a constant rate.	





Concept: Cell Specialization & Development (13.2, 13.3)

Key Definitions

Differentiation: The process where cells from meristems mature to perform specific functions, involving major structural changes (e.g., a cell losing protoplasm to form a tracheary element).

Dedifferentiation: The phenomenon where living, differentiated cells regain the capacity to divide under certain conditions (e.g., formation of interfascicular cambium from parenchyma).

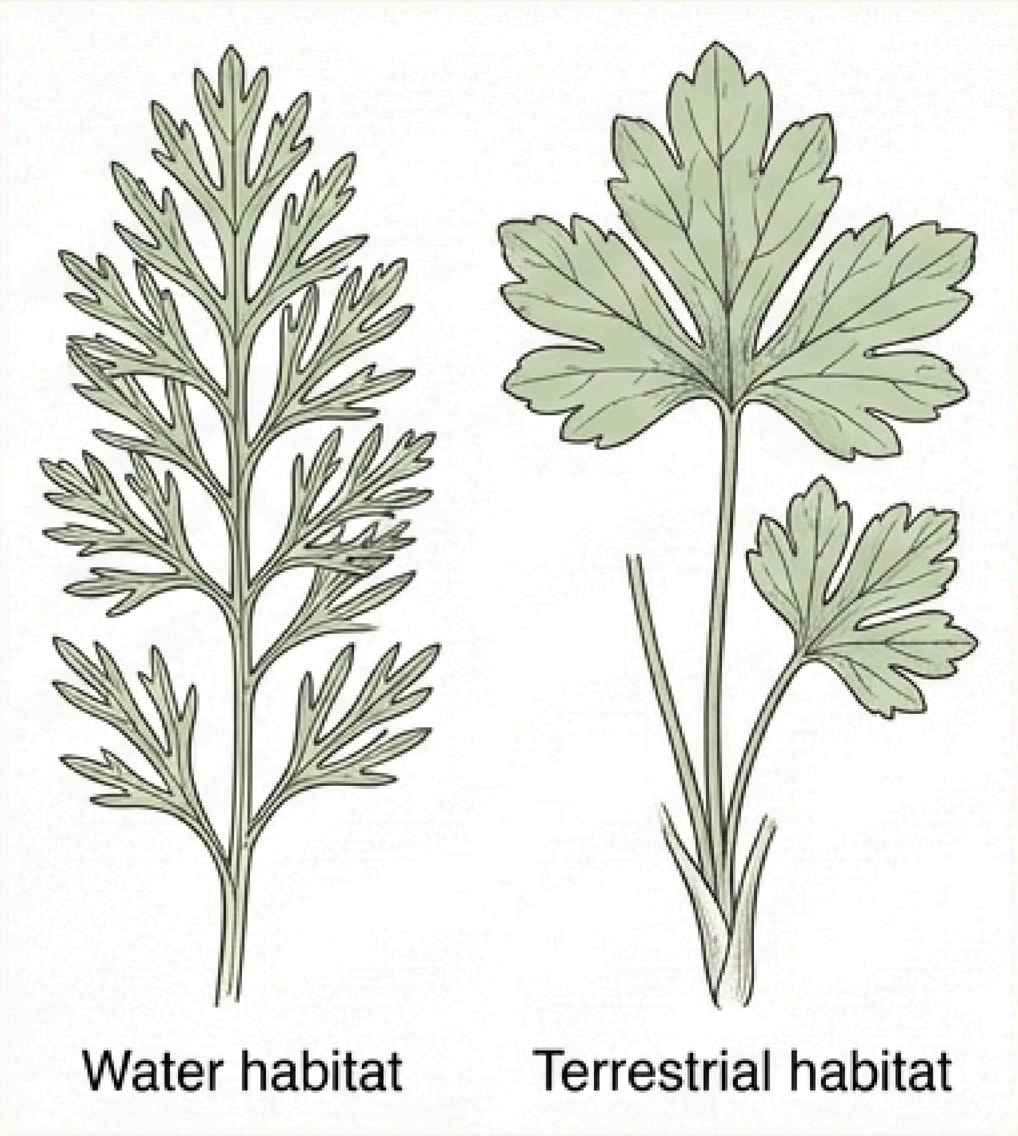
Redifferentiation: When dedifferentiated cells divide and their products mature to perform specific functions, losing the ability to divide again (e.g., formation of secondary xylem).

Development: The sum of all changes in a life cycle, from seed germination to senescence.
Development = Growth + Differentiation.

Plasticity: The ability of plants to follow different pathways in response to the environment or life phases to form different structures.

Example of Plasticity: Heterophylly

- **In Cotton, Coriander, Larkspur**: Leaves of the juvenile plant are different in shape from those in mature plants.
- **In Buttercup**: The shape of leaves produced in air is different from from leaves produced in water.



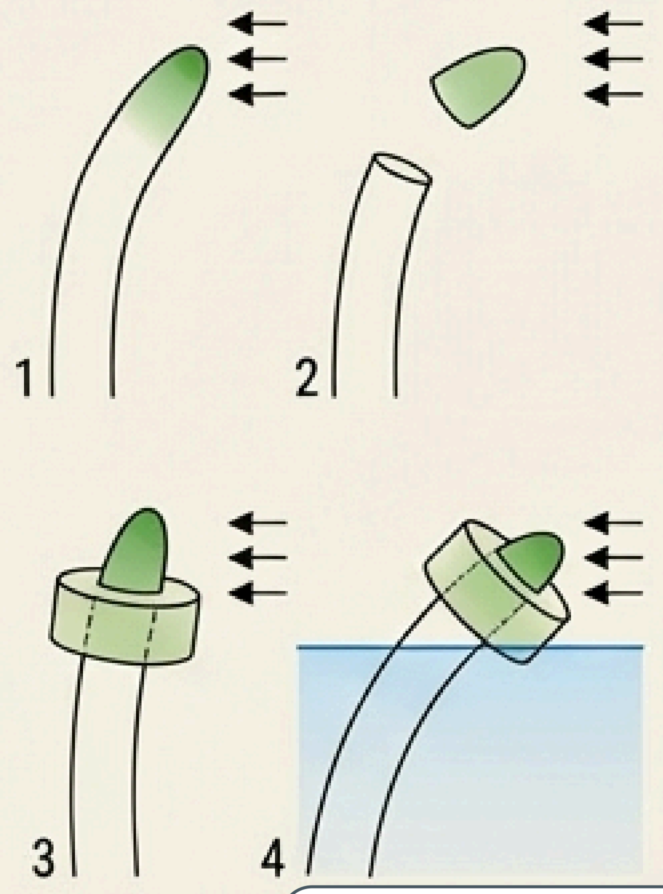
Concept: Plant Growth Regulators - Introduction (13.4.1, 13.4.2)

Characteristics & Functional Groups

- **Characteristics:**
  PGRs (or phytohormones) are small, simple molecules of diverse chemical composition that regulate plant growth.
- **Chemical Nature:** Indole compounds (IAA), adenine derivatives (kinetin), carotenoid derivatives (ABA), terpenes (GA_3), and gases (ethylene).
- **Functional Groups:**
  - **Plant Growth Promoters:** Involved in cell division, enlargement, flowering, fruiting (e.g., Auxins, Gibberellins, Cytokinins).
  - **Plant Growth Inhibitors:** Involved in dormancy, abscission, and stress responses (e.g., Abscisic Acid). Ethylene fits both groups but is largely an inhibitor.

The Discovery of PGRs

Scientist(s)	Observation / Source	PGR Group
Charles & Francis Darwin	Coleoptiles of canary grass bent towards light (phototropism).	**Auxin**
F.W. Went	Isolated the 'influence' from the tips of oat seedlings.	**Auxin**
E. Kurosawa	Found that sterile filtrates of the fungus *Gibberella fujikuroi* caused 'bakanae' disease in rice.	**Gibberellin**
F. Skoog & Miller	Identified a cytokinesis-promoting substance, termed kinetin, from autoclaved herring sperm DNA.	**Cytokinin**
H.H. Cousins	Confirmed a volatile substance from ripened oranges hastened ripening in bananas.	**Ethylene**
Independent Researchers	Identified three inhibitors (inhibitor-B, abscission II, dormin) which were chemically identical.	**Abscisic Acid (ABA)**



Concept: Physiological Effects of Promoters (13.4.3.1, 13.4.3.2)

Auxins ('to grow')

Types: Natural (IAA, IBA), Synthetic (NAA, 2,4-D).
Produced in: Growing apices of stems and roots.

Key Functions & Applications:

- Apical Dominance: Apical bud inhibits lateral bud growth. Decapitation is used in tea plantations and hedge-making.
- Initiates rooting in stem cuttings.
- Initiates rooting in stem cuttings.
- Promotes flowering (e.g., pineapples).
- Induces parthenocarpy (e.g., tomatoes).
- Prevents early fruit/leaf drop but promotes abscission of older ones.
- Herbicide: 2,4-D kills dicot weeds without affecting mature monocots.
- Controls xylem differentiation.

Gibberellins (GA)

Types: Over 100 types (GA_1, GA_2, etc.); all are acidic. GA_3 is the most studied.

Key Functions & Applications:

- Increases length of grape stalks and elongates apples.
- Delays senescence, extending the market period for fruits.
- Speeds up the malting process in the brewing industry.
- Increases sugarcane yield (up to 20 tonnes/acre) by increasing stem length.
- Promotes early seed production in juvenile conifers.
- Bolting: Promotes internode elongation prior to flowering in rosette plants (beet, cabbages).

Concept: Physiological Effects of Other PGRs (13.4.3.3 – 13.4.3.5)

Cytokinins	Ethylene (Gaseous PGR)	Abscisic Acid (ABA)
Primary Effect: Promotes Cytokinesis (cell division).	**Primary Effect:** Fruit ripening; largely inhibitory.	**Primary Effect:** Growth inhibition and stress response.
Discovery/Types: Kinetin (from herring sperm DNA, not natural in plants). Zeatin (natural, from corn kernels).	**Synthesized in:** Tissues undergoing senescence and ripening fruits.	**Also known as:** The Stress Hormone.
Functions: • Helps overcome apical dominance. • Promotes lateral shoot growth. • Promotes nutrient mobilisation, which delays leaf senescence. • Helps produce new leaves and chloroplasts.	**Functions:** • Highly effective in fruit ripening; enhances respiration (**respiratory climactic**). • Promotes senescence and abscission of leaves/flowers. • Breaks seed and bud dormancy. • Promotes rapid internode/petiole elongation in deep water rice plants. • Promotes root growth and root hair formation.	**Functions:** • Inhibits seed germination. • Stimulates the closure of stomata during water stress. • Plays a key role in seed development, maturation, and dormancy. • Acts as an antagonist to Gibberellins (GAs).

Concept: Direct NCERT Lines & NEET Pickups

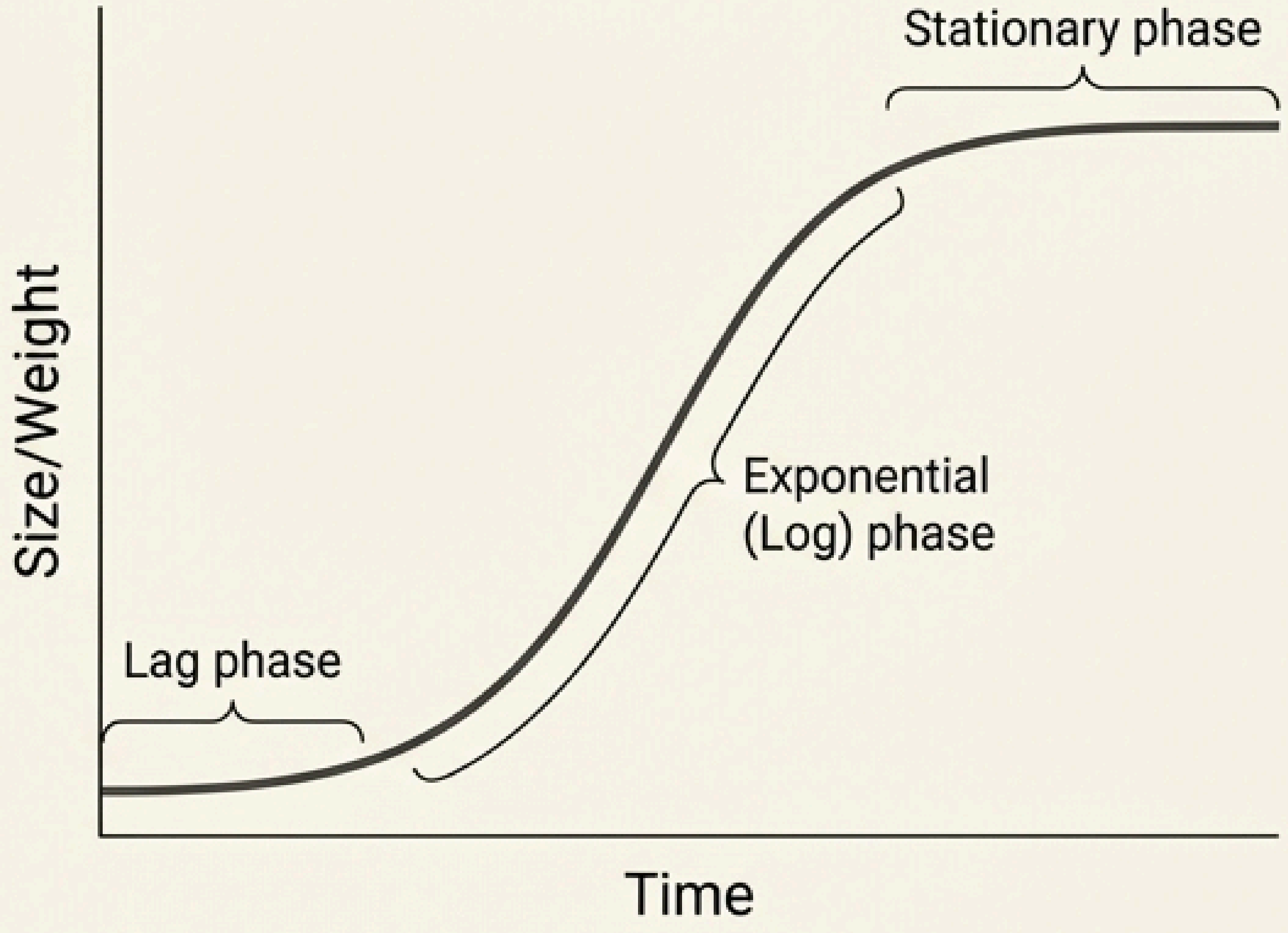
⚠ NCERT Line

"Growth can be defined as an **irreversible permanent increase in size** of an organ or its parts or even of an individual cell."

★ Direct NEET Pickup

- "This form of growth wherein new cells are always being added to the plant body by the activity of the meristem is called the **open form of growth**."
- "A **sigmoid curve** is a characteristic of living organism growing in a natural environment."
- "In most higher plants, the growing apical bud inhibits the growth of the lateral (axillary) buds, a phenomenon called **apical dominance**."
- "**2, 4-D**, widely used to kill **dicotyledonous weeds**, does not affect mature **monocotyledonous plants**."
- "Gibberellins also promotes **bolting** (internode elongation just prior to flowering) in beet, cabbages and many plants with rosette habit."
- "This rise in rate of respiration [during fruit ripening] is called **respiratory climactic**."
- "ABA stimulates the closure of stomata… Therefore, it is also called the **stress hormone**."

Idealised Sigmoid Growth Curve



Examiner's Focus:

- Identification of the three phases.
- The formula for the exponential phase: $\mathbf{W_1 = W_0e^{rt}}$.
- Understanding this curve is typical for growth in a natural environment with limited resources.

Apical Dominance in Plants

Examiner's Focus:

- The role of **Auxin** from the apical bud in suppressing lateral growth.
- The effect of **decapitation** (removal of shoot tips).
- Practical applications: **tea plantations** and **hedge-making**.

Comparative Analysis of PGR Functions

Function	Auxin	Gibberellin (GA)	Cytokinin	Ethylene	Abscisic Acid (ABA)
Apical Dominance	**Maintains**	-	**Overcomes**	-	-
Stem Elongation	Promotes (cell enlarg.)	**Promotes (Bolting)**	-	Promotes (deep water rice)	**Inhibits**
Cell Division	Promotes	-	**Strongly Promotes**	-	**Inhibits**
Seed Dormancy	-	**Breaks**	-	Breaks	**Induces**
Fruit Ripening	-	-	-	**Strongly Promotes**	-
Senescence & Abscission	Promotes (older organs)	**Delays**	**Delays**	Promotes	Promotes
Stomatal Closure	-	-	-	-	**Induces (Stress)**

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Exam-Critical Points: Plant Growth & Development

Previous Year NEET Question Themes

- **PGR Functions**: Direct matching of a PGR to its specific function (e.g., GA and bolting; Ethylene and respiratory climactic; Auxin and apical dominance).
- **Agricultural Applications**: Which PGR is used as a herbicide (2,4-D), for rooting cuttings (Auxin), or to increase sugarcane yield (GA)?
- **Discovery**: Matching scientists to their respective discoveries (e.g., F.W. Went, Kurosawa).
- **Growth Concepts**: Questions on the sigmoid curve, plasticity, and differentiation.

Common Traps & Misconceptions

- **Auxin's role in Abscission**: Remember it **PREVENTS** drop of young leaves/fruits but **PROMOTES** drop of older ones.
- **Kinetin vs. Zeatin**: Kinetin does **not** occur naturally in plants; it was discovered from autoclaved herring sperm DNA. Zeatin is a natural cytokinin.
- **Ethylene's Dual Role**: While largely an inhibitor (promotes senescence), it also promotes growth in specific cases (e.g., petiole elongation in deep water rice).

One-line Memory Hooks

- To overcome apical dominance: "Cut the **Auxin** head, let the **Cytokinin** sides grow."

Conceptual Application & Analysis

Assertion (A) – Reason (R) Logic

A: Spraying sugarcane crops with gibberellins **increases the yield** of sugar.
R: Gibberellins **increase the length of the stem** in sugarcane plants.

Analysis: Both A and R are true, and R is the correct explanation of A. Sugar is stored in the stem, so a longer stem means a higher yield.

A: Abscisic acid (ABA) is called the **stress hormone**.
R: ABA **promotes the closure** of stomata during conditions of water stress.

Analysis: Both A and R are true, and R is the correct explanation of A. Stomatal closure is a primary mechanism to **conserve water** under stress.

True/False Style Conceptual Checks

Statement I: Kinetin is a naturally occurring cytokinin isolated from corn kernels.
Statement II: 2,4-D is a synthetic auxin widely used to kill monocotyledonous weeds.

Analysis: Statement I is **False** (**Zeatin** was isolated from corn kernels; Its isolated from corn kernels; , Kinetin is **not natural** in plants). Statement II is **False** (2,4-D is used to kill **dicot weeds**).

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Plant Growth & Development: High-Yield Revision

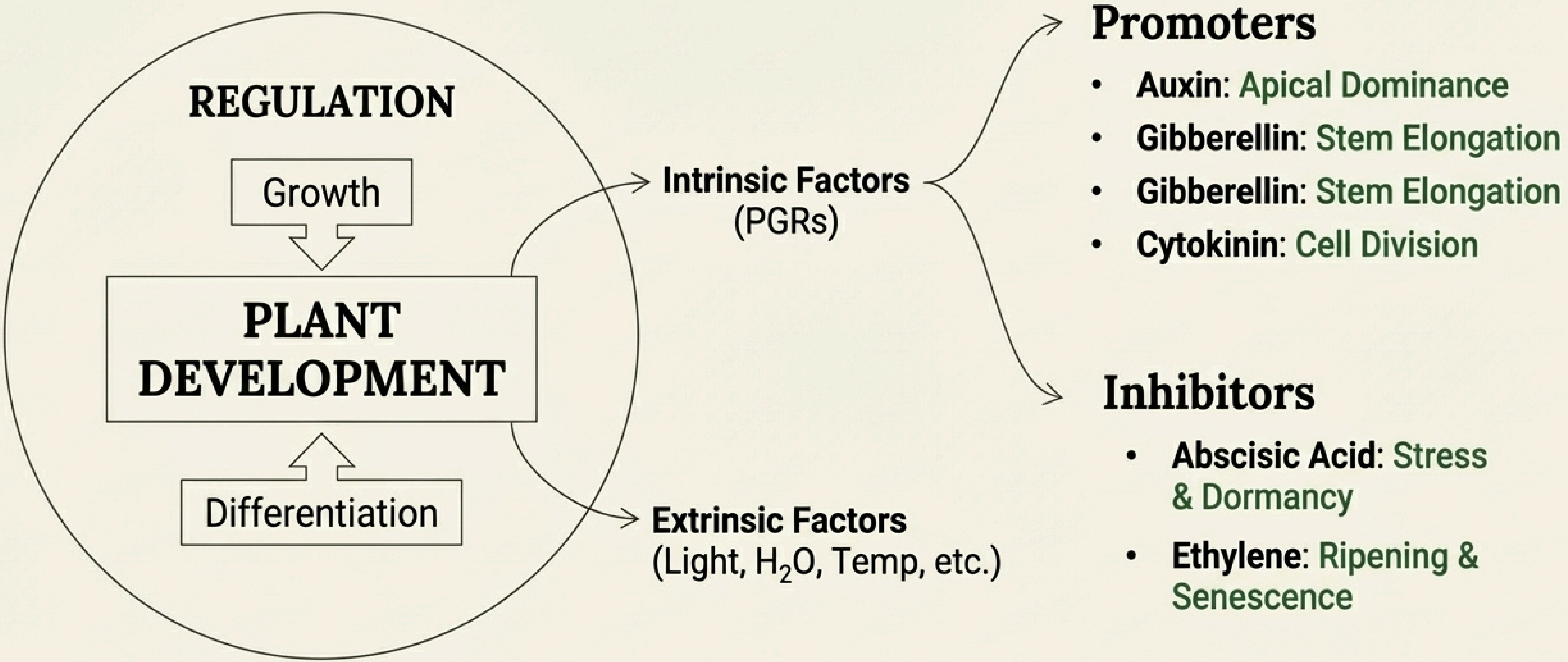
Bullet-Point Summary

- **Growth**: Irreversible, indeterminate (due to meristems), open form. Measured in size, number, weight, etc.
- **Phases**: Meristematic (division), Elongation (enlargement), Maturation (specialization).
- **Rates**: Arithmetic (Linear, $L_t=L_0+rt$), Geometric (Sigmoid, $W_1=W_0e^{rt}$).
- **Cell States**: Differentiation → Dedifferentiation → Redifferentiation.
- **Development**: Growth + Differentiation. Influenced by intrinsic (PGRs) & extrinsic factors.
- **Plasticity**: Environmental or developmental heterophylly (e.g., buttercup, larkspur).
- **Auxin**: Apical dominance, rooting, parthenocarpy.
- **Gibberellin**: Bolting, stem elongation, breaks dormancy.
- **Cytokinin**: Cell division, overcomes apical dominance, delays senescence.
- **Ethylene**: Fruit ripening (respiratory climactic), senescence, abscission.
- **ABA**: Stress hormone, stomatal closure, induces dormancy.

NCERT Keywords

Indeterminate Growth | Open Form of Growth | Meristem | Sigmoid Curve | Efficiency Index | Plasticity | Heterophylly

Differentiation | Apical Dominance | Bolting | Senescence | Parthenocarpy | Respiratory Climactic | Stress Hormone



'If you remember only this much, you can solve most questions'

Development in plants is the sum of growth and differentiation. This entire process is controlled by intrinsic factors, primarily the five groups of PGRs which act as promoters or inhibitors, and extrinsic environmental factors. Understanding the specific, key roles of each PGR is the key to mastering this chapter.